

transistor T1 conducts. The common terminal of IC1 connects to the ground and IC1 outputs 6V. As a result, transistor T2 does not conduct and the relay remains de-energised. The light bulb remains 'off' as the mains connection is not completed through the relay contacts.

During nighttime, when no light falls on LDR1, it offers a high resistance at the base junction of transistor T1. So the bias is greatly reduced and

T1 doesn't conduct. Effectively, this removes the common terminal of IC1 from ground and it directs the full input DC to the output. Transistor T2 conducts and the relay energises to light up the bulb as mains connection completes through the relay contacts.

As LDR1 is in parallel to VR1+R3 combination, it effectively applies only half of the total resistance of the network formed by R3, VR1 and LDR1 to the junction at T1 in total

darkness. In bright light, it greatly reduces the total effective resistance at the junction.

The circuit is simple and can be assembled on a small general-purpose PCB. Use a heat-sink for IC1. Make sure that LDR1 and the light bulb are well separated.

The circuit can be used for street-lights, tubelights or any other home electrical lighting system that needs to be automated. ●

SPY EAR

■ T.K. HAREENDRAN

What binoculars do to improve your vision, this personal sound enhancer circuit does for listening. This lightweight gadget produces an adjustable gain on sounds picked up from the built-in high-sensitivity condenser microphone. So you can hear what you have been missing. With a 6V (4×1.5V) battery, it produces good results.

As shown in Fig. 1, a small signal amplifier is built around transistor BC547 (T1). Transistor T1 and the related components amplify the sound



signals picked up by the condenser microphone (MIC). The amplified signal from the preamplifier stage is fed to input pin 3 of IC LM386N (IC1) through capacitor C2 (100nF) and volume control VR1 (10-kilo-ohm log). A decoupling network comprising resistor R5 and capacitor C3 provides the preamplifier block with a clean supply voltage.

Audio amplifier IC LM386N (IC1) is designed for operation with power

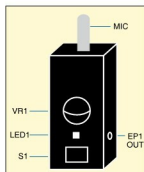


Fig. 2: Compact unit of spy ear

supplies in the 4-15V DC range. It is housed in a standard 8-pin DIL package, consumes very small quiescent current and is ideal for battery-powered portable applications.

The processed output signal from capacitor C2 goes to one end of volume control VR1.

The wiper is taken to pin 3 of LM386N audio output amplifier. Note that the R6-C4 network is used to RF-decouple positive-supply pin 6 and R8-C7 is an optional Zobel network that ensures high frequency stability when feeding an inductive headphone load.

Capacitor C6 (22μF, 16V) wired between pin 7 and ground gives additional ripple rejection. The output of LM386N power amplifier can safely drive a standard 32-ohm monophonic headphone/earphone.

Assemble the circuit on a small general-purpose PCB and house in a suitable metallic enclosure with an integrated battery holder and headphone/earphone socket as shown in Fig. 2. Fit the on/off switch (S1), volume control (VR1) and power indicator (LED1) on the enclosure. Finally, fit the condenser microphone (MIC) on the front side of the enclosure and link it to the input of the preamplifier via a short length of the shielded wire. ●

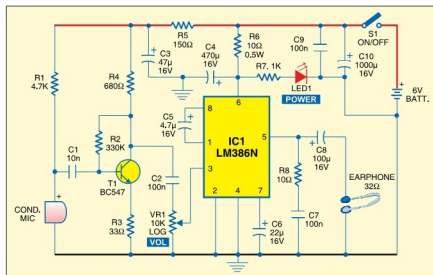


Fig. 1: Circuit for spy ear